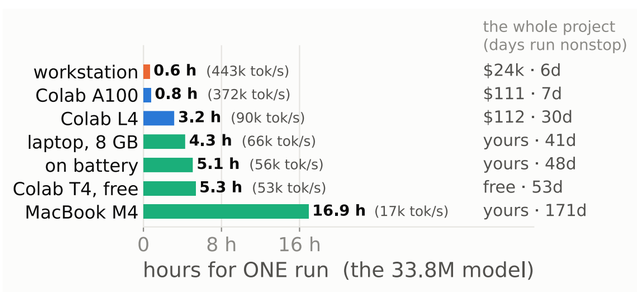
UNIVERSITY of WASHINGTON

CSED 505: BUILDING A MODEL FACTORY

501 the statistics · 502 the mechanics · 503 the language stack · 504 scale — All four end when a model finishes training. None covers needing a hundred models and having to believe the differences between them.

Jeffrey Stall · Patrick Kwok · Leon Wan — CSED 504. This board is the machinery; the upper board is the experiment it served.

You do not need the workstation



An **8 GB laptop GPU beats the free tier, even unplugged**. The two paid Colab tiers bill the same — **$111 for 7 days** against **$112 for 30** — so the faster tier buys time, not money. Electricity for all 148 GPU-hours was **$7**; the workstation that ran them, **$24,000**.

When memory is a wall, and when it is only a speed limit

What has to **fit** is the model. Everything above that is a choice.

The floor is the model — weights, gradients, optimizer state: 12–16 bytes a parameter. Our 98M: **1.6 GB**. A 7B model: **84–112 GB**, which no consumer card has. *(arithmetic)*

Everything above it is batch, and batch is optional. Accumulation runs a 16,384-token step as smaller passes — same update, **within 2%** of the cost.

So an 8 GB laptop trains the step a 96 GB card trains. A 10 GB run folds into **6.1 GB at 28,027 tok/s**.

Nobody on the chart beside this needed 80 GB. 96, 80, 24, 16, 8 GB — all ran the same step. The A100's price buys its **speed**, not capacity.

More memory does not buy speed. Batch 2048 uses **89.7 GB** and is *slower* than 128.

And Windows does not warn you. An oversized batch spills to system RAM: **5,075 tok/s, no error**.

The factory, in numbers

197	pretraining runs, every one from scratch
892	fine-tuning runs on top of them, across two tasks
17	languages measured, 12 of them pretrained from scratch
148	GPU-hours, two cards, 71 kWh at the wall
62,500	steps a run — 1.024B tokens of updates, the unit that makes two machines comparable
0	identity collisions, because a vocabulary carries a hash

What we made faster

2.07× on the four-cell benchmark — only **1.32×** of it efficiency; the rest is a second card nobody had used. **A third of the twenty were rejected**, and a list of only the wins would be marketing rather than a log. Every row had to make the *same* experiment faster.

WHAT CHANGED	MEASURED	KEPT?
THE DATA PATH		
1 · dataset on the card, no loader	13.8k → **18.5k img/s**	yes
2 · images as uint8, cast per batch	4× less memory, free cast	yes
3 · augment on the GPU	no PIL, no host copy	yes
4 · tokenize once, up front	53 s vs 85 min — **96×**	yes
5 · token dtype from the vocabulary	16k → **2 bytes a token**	yes
PRECISION AND KERNELS		
6 · bf16 autocast, not fp16	**1.34×** on sm_89, no loss-scale	yes
7 · channels_last for CNNs, bf16 only	fp16 + it is **3.5× slower**	guarded
8 · channels_last for ViTs	a no-op on a transformer	no
9 · TF32 for matmul and cuDNN	free on Ampere and later	yes
10 · cudnn.benchmark = True	worth it when shapes are fixed	yes
11 · fused SGD and AdamW	one kernel, not a chain	yes
12 · torch.compile	no Triton on Windows	can't
13 · attention backend	already sdpa	no change
HOST-SYNC DISCIPLINE		
14 · drop the per-step .item(), CIFAR	**~7%**	yes
15 · the same at MLM scale	**2%** — a live loss is worth it	no
16 · zero_grad(set_to_none=True)	skips a full zero-fill	yes
BATCHING AND SCHEDULING		
17 · batch 64 → 128	**1.31×** — 2048 is *slower*	yes
18 · checkpointing for speed	peak is **3.3 GB of 96**	no
19 · use the second card	**91% / 93%** busy at 300 W	yes
20 · longest-budget-first queue	**2.55×** projected, 2.07 seen	yes

Budget in tokens, not epochs

In 502–504 epochs were correct — the dataset was fixed. Here it is the experiment. Our ladder spans 4M to 1,024M tokens: "40 epochs each" hands the big arm **256× the compute**.

So fix tokens, and let epochs fall out. Every run is **1.024B tokens of updates** — 62,500 steps was never chosen, it is 1.024B ÷ a 16,384-token batch.

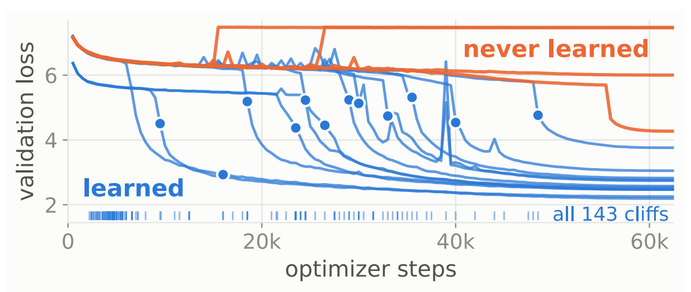
Tokens ÷ tok/s = the wall clock. 1.024B ÷ 443k tok/s ≈ **38 min** — the hardware chart's top bar, and how a promise gets measured.

Fixed compute makes the ladder readable. The 4M corpus is seen **256 times**; the 1,024M corpus, **exactly once**. Identical compute — only the data moves.

And a truncated run is not a cheaper run. The schedule anneals to zero at the *planned* end — a run stopped halfway is mid-schedule, a different experiment.

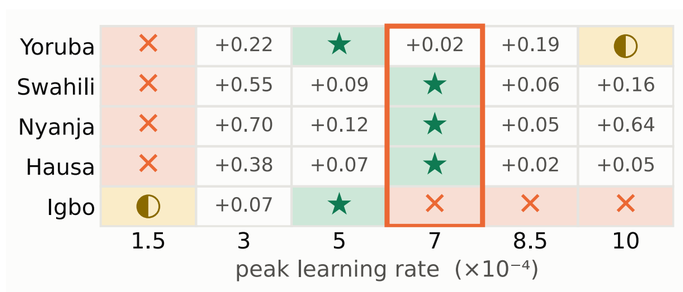
Our own records carried the bug. A field named epoch reads **125** where the truth is **16 passes** — it kept the name and lost the meaning.

You cannot judge a run early



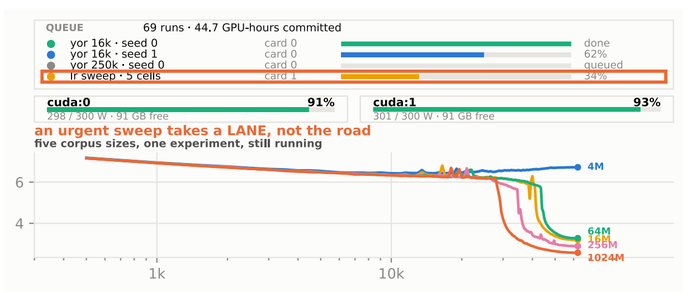
Yoruba, English and ten more, all on one axis. Every run that learns sits flat and then falls off a cliff — and **no two fall at the same step**: earliest **2,200**, latest **48,500**. So "still flat at step k" is evidence of nothing. **35 of 197 never learned.**

The setting four languages want destroys the fifth



Five languages, six learning rates, sixty runs, each cell scored against **that language's own best**: ★ its winner, × far behind it, ◐ one seed of two. **Four land at their best in the outlined column. Igbo collapses there** — the same setting, a wasted night.

One screen, both cards, a rush order at 9 p.m.



The chart is five English corpus sizes separating while they run. On 9 August **44.7 GPU-hours sat across 69 runs** and Patrick needed a sweep: a fleet pins to one card, so it took **card 1** while the queue kept card 0. **Twenty minutes** to running, on a measured promise.

An identity is everything that changes the answer

Two runs silently overwrote each other — same filename, different vocabularies, and neither record said so. A filename is an identity, so the vocabulary carries a **hash** rather than a label: 15abd33de5af. The hash also caught two people who prepared **"the same" corpus** and built different vocabularies. **197 pretraining and 892 fine-tuning runs later, no collision.**

A result has to beat your own noise

Run the same recipe twice and it lands in two different places. That is the seed, not the change you made. So we measured **our own noise first** and made every claim clear it by **2.27×** — at three runs a side, no test can call anything significant. The spread is a property of **each cell** — re-measured, never a constant we reused. **Two of our own claims did not clear it. We retired them.**

What we do not claim, and what we would do next

Nobody on this team reads Yoruba — every judgement here is a benchmark number. The corpus is scraped web text nobody consented to. The audit gate prints **9 claims: 6 supported, 2 not, 1 underpowered**. Next: a wall meter, a corpus audit, the evaluation we could not do.

A factory that makes one thing is not a factory

English is the ruler. *Why:* the one language where data was never the constraint. *Learned:* **more data stops helping past 64M tokens** — a fact Yoruba could never establish, having no more.

French, Indonesian, Mandarin are the control. *Why:* XLM-R knows these three well, killing the objection that it is just "unfamiliar languages". *Learned:* they cost **1.04, 1.01, 0.95** — the penalty tracks *coverage*, not "African".

Twelve African languages are the gradient. *Why:* one language is an anecdote; seventeen make a slope. *Learned:* the split falls exactly along coverage — **Wolof, uncovered yet cheap at 1.31**, is the exception an anecdote hides.

Five languages got the whole sweep. *Why:* "adding a language is one function call" is only true if the settings transfer too. *Learned:* they do not — **the rate four languages want destroys Igbo.**

None of us speaks any of them. So the spread is the evidence, not any one score. The factory made the spread affordable: **nine functions** are its whole interface, so adding a language is one call — **twelve pretrained in 48 minutes**.

Sources

MODELS

A. Conneau et al., "Unsupervised cross-lingual representation learning at scale" (XLM-R), ACL 2020

M. Marone et al., "mmBERT," arXiv:2509.06888

Y. Liu et al., "RoBERTa," arXiv:1907.11692

J. Devlin et al., "BERT" — its 80/10/10 masking, NAACL 2019

K. Ogueji et al., "Small data? No problem!" (AfriBERTa), EMNLP MRL 2021

DATA

D. I. Adelani et al., "SIB-200," EACL 2024

D. I. Adelani et al., "MasakhaNER 2.0," EMNLP 2022

G. Penedo et al., "FineWeb2," arXiv:2506.20920

METHOD

L. N. Smith, one-cycle schedules, arXiv:1803.09820

J. Dodge et al., "Show your work," EMNLP 2019 — no result without the search budget that produced it

AI statement. We used Claude Code throughout to turn ideas into working code. It made the building faster and the verifying no faster at all — every number on this board is regenerated from committed records by a gate that fails the build when a claim and a record disagree. · Built with PyTorch and HuggingFace tokenizers, plus NumPy and matplotlib; no training framework above them — the loop, the masking, the scheduler and the run records are this project's own. Long jobs ran on a 2 × RTX PRO 6000 Blackwell workstation, benchmarked against Colab A100 / L4 / T4, a MacBook Pro (M4 Pro, MPS) and an 8 GB laptop. · Nineteen references, in full in report 09 §References. · Source: reports/12-the-bottom-board.md · every number on this board regenerates from committed runs/ records